

In T.208/84 (OJ 1987, 14) the method Claims 1-7 and 12 were directed to methods for digitally processing images. One basic issue to be decided was whether or not such a method was excluded from patentability under Art. 52(2) and (3) EPC 1973 on the ground that it was a mathematical method as such. The board noted that there could be little doubt that any processing operation on an electric signal can be described in mathematical terms. The characteristic of a filter, for example, can be expressed in terms of a mathematical formula. A basic difference between a mathematical method and a technical process can be seen, however, in the fact that a mathematical method or a mathematical algorithm is carried out on numbers (whatever these numbers may represent) and also provides a result also in numerical form, the mathematical method or algorithm being only an abstract concept prescribing how to operate on the numbers. No direct technical result is produced by the method as such. In contrast, if a mathematical method is used in a technical process, that process is carried out on a physical entity (which may be a material object but equally an image stored as an electric signal) by some technical means implementing the method and provides as its result a certain change in that entity. The technical means might include a computer comprising suitable hardware or an appropriately programmed general purpose computer. The board was, therefore, of the opinion that, even if the idea underlying an invention may be considered to reside in a mathematical method, a claim directed to a technical process in which the method is used does not seek protection for the mathematical method as such. In contrast, a "method for digitally filtering data" remains an abstract notion not distinguished from a mathematical method as long as it is not specified what physical entity was represented by the data and it forms the subject of a technical process, i.e. a process which is susceptible of industrial application (see also T.1161/04, T.212/94).

In T.953/94, claim 1 of the main request related to a method of generating with a digital computer a data analysis of the cyclical behaviour of a curve represented by a plurality of plots relating two parameters to one another (use of mathematical method in physical process). The board held that such a method could not be regarded as a patentable invention, because an analysis of the cyclical behaviour of a curve was clearly a mathematical method excluded as such from patentability. The reference to a digital computer only had the effect of indicating that the claimed method was carried out with the aid of a computer, i.e. a programmable general-purpose computer, functioning under the control of a program excluded as such from patentability. The fact that the description disclosed examples in both non-technical and technical fields was confirmation that the problem solved by the claimed mathematical method was independent of any field of application and could thus lie, in the case at issue, only in the mathematical and not in a technical field.

In T.27/97, the appellant (opponent) interpreted the claim to mean that its subject-matter, despite the statement that a method for use in electronic systems was involved, was confined to purely intellectual methods and thus excluded by Art. 52(2)(c) EPC 1973. The board disagreed, ruling that, according to claim 1, the application claimed a method, for use in electronic systems, of encrypting or decrypting a message (represented in the form of a digital word using RSA-type public-key algorithms). So the invention was clearly a method in the computer and telecommunications field and thus not excluded under

Art. 52(2) and (3) EPC 1973, even if based on an abstract algorithm or mathematical method.

In T 1326/06 the board took the view that processes for encoding/decoding or signing electronic communications by RSA had to be regarded as technical processes, even if they were based essentially on mathematical processes (see also decisions T 953/04, point 3.3 of the Reasons and T 27/97, point 3 of the Reasons).

In T 1784/06 (point 3.1.1 of the Reasons) the board stated that the algorithm is a mathematical (inter alia Boolean) method and mathematical methods as such are deemed to be non-inventions (Art. 52(2) and (3) EPC). A technical character of the algorithm could be recognised only if it served a technical purpose (see e.g. T 1227/05, point 3.1 of the Reasons, OJ 2007, 574) (see also T 306/10).

In T 556/14 the invention related to a method for masking a private key used in cryptographic operations on a security token such as a smartcard against power analysis attacks. The board held that due to the express reference to a smart card on which the key parts and also the new parts are stored, the claimed method of masking was not a mathematical method as such.

2.3. Aesthetic creations

Subject-matter relating to aesthetic creations will usually have both technical aspects, e.g. a 'substrate' such as a canvas or a cloth, and aesthetic aspects, the appreciation of which is essentially subjective, e.g. the form of the image on the canvas or the pattern on the cloth. If technical aspects are present in such an aesthetic creation, it is not an aesthetic creation 'as such' and it is not excluded from patentability. The aesthetic effect itself is not patentable, neither in a product nor in a process claim. Nevertheless, if an aesthetic effect is obtained by a technical structure or other technical means, although the aesthetic effect itself is not of a technical character, the means of obtaining it may be (Guidelines G-II, 3.4 – March 2022 version).

In T 686/90 the board was called upon to decide whether the feature "work of art in the style of stained glass" meant that it was excluded from patentability under Art. 52(2)(b) EPC 1973. The board held that functional information referring to general aesthetic creations did not define an aesthetic creation as such, at least provided that and insofar as such information adequately identified technical features of the subject-matter of the claim. Since an aesthetic creation (not formally specified) as the stated purpose, together with the other features, adequately defined a technical subject-matter in the claim, there was no aesthetic creation as such. For this reason there could be no objection to the claim under Art. 52(2)(b) EPC 1973 on the basis of Art. 52(3) EPC 1973.

In decision T 119/88 (OJ 1990, 395) the subject-matter of the application in question related to a flexible disk jacket made of a plastic sheet presenting to the outside world a surface colour of a certain minimum light intensity. The board first of all stated that the feature of having a specific colour as such did not constitute a technical feature indicating that an object or device was entirely or partly covered by that colour; however, the board

did not rule out the possibility that this did not hold in all circumstances. The feature taken by itself might not seem to reveal any technical aspect, but its technical or non-technical character could be decided by **the effect it brought about after being added to an object** which did not comprise the feature before (see e.g. T 336/14). In considering whether the subject-matter of a claim is excluded from patentability under Art. 52(2) and (3) EPC 1973, i.e. whether it is non-technical or technical, that claim has to be considered in its entirety. This is in line with decisions T 38/86 and T 65/86, in which Art. 52(3) EPC 1973 was interpreted to mean that the subject-matter of a claim is not excluded from patentability under Art. 52(2)(c) EPC 1973 if the said subject-matter contributes anything to the prior art in a field that is not excluded from patentability. In these decisions this interpretation was only applied with regard to Art. 52(2)(c) EPC 1973. However, once applied to Art. 52(2)(c) EPC 1973 it must hold for all the paragraphs of Art. 52(2) EPC 1973, since Art. 52(3) EPC 1973 refers to all the items summed up in Art. 52(2) EPC 1973. The board stated that it had to be investigated whether the effects implied by the sole feature of possessing a specific colour, could make that feature into a feature not excluded from patentability the known object or device provided with that colour. The board concluded that the alleged resistance to fingerprints was a purely aesthetic effect which contributed nothing technical to the invention concerned (Art. 52(2)(b) EPC 1973) and the advantage of easy classification by colour represented a non-technical effect in the form of a presentation of information. As such it was excluded from patentability under Art. 52(2)(d) and (3) EPC 1973 (see also T 962/91).

In T 617/11 the appellant I (opponent I) considered that there was no objective technical problem to be solved since the alleged obtained effects related to aesthetic effects. It argued that features could not be regarded as technical and justify patentability if their effects related to the subjective senses of the viewer (T 119/88, OJ 1990, 395) and concluded that, since all the features of claim 1 related to non-technical effects of colours and shapes, the claimed container could not be regarded as a patentable invention (Art. 52(2)(b) and (d) EPC). The board noted that the case law of the boards of appeal in relation to the technicality of inventions and the exclusions from patentability as provided in Art. 52(2) EPC was summarised in T 154/04 (OJ 2008, 46) for inventions which have a mix of technical and non-technical features. The fact that the non-technical features provide the only contribution over the known prior art and thus lead to the conclusion of non-patentability under Art. 52(2) EPC, as held in T 119/88 (supra), had clearly been overruled by the large body of jurisprudence on this issue, as summarised in T 154/04 and established by the decisions citing T 154/04 for this aspect. Hence, the board could not find fault with the reasoning and the findings of the opposition division.

For more information on the inventive-step analysis in connection with features relating to subject-matter excluded under Art. 52(2) EPC, see chapter I.D.9.2.2 "Excluded subject-matter".

2.4. Computer-implemented inventions

2.4.1 Definition of computer-implemented invention

"Computer-implemented invention" is an expression intended to cover claims which involve computers, computer networks or other programmable apparatus wherein at least one feature is realised by means of a computer program (Guidelines G-II, 3.6 – March 2022 version).

2.4.2 Distinction from computer programs

A computer program and a corresponding computer-implemented method are distinct from each other. The former refers to a sequence of computer-executable instructions specifying a method while the latter refers to a method being actually performed on a computer (Guidelines G-II, 3.6 – March 2022 version).

2.4.3 Scope of Article 52(2)(c) EPC exception to patentability

Art. 52(2)(c) EPC states that programs for computers shall not be regarded as inventions within the meaning of Art. 52(1) EPC and are therefore excluded from patentability.

Art. 52(3) EPC establishes an important limitation to the scope of this exclusion. According to this provision, the exclusion applies only to the extent to which a European patent application or a European patent relates to programs for computers "as such". In T 935/97 the board stated that the combination of the two provisions (Art. 52(2) and (3) EPC 1973) demonstrates that the legislators did not want to exclude from patentability all programs for computers. In other words, the fact that only patent applications relating to programs for computers as such are excluded from patentability means that patentability may be allowed for patent applications relating to programs for computers where the latter are not considered to be programs for computers as such (see also T 1924/17, points 17 to 19.4 of the Reasons, on the background to Art. 52 EPC).

Claims directed to a computer-implemented method, a computer-readable storage medium or a device cannot be objected to under Art. 52(2) and (3) EPC as any method involving the use of technical means (e.g. a computer) and any technical means itself (e.g. a computer or a computer-readable storage medium) have technical character and thus represent inventions in the sense of Art. 52(1) EPC (T 258/03; T 424/03; G 3/08, OJ 2011, 10).

2.4.4 Patentability of computer programs: G 3/08

The President of the EPO, making use of her power under Art. 112(1)(b) EPC, made a referral to the Enlarged Board of Appeal on several points of law relating to the application of the exclusion of computer programs as such, and the limits of patentability in the field of computing. In particular, the President of the EPO referred to the Enlarged Board of Appeal the following point of law (see G 3/08, Question 1 of the referral), "Can a computer

program only be excluded as a computer program as such if it is explicitly claimed as a computer program?"

In Opinion G 3/08 (OJ 2011, 10), the Enlarged Board of Appeal held that a claim utilising a synonym for "computer program", such as "a sequence of computer-executable instructions" or "an executable software module" perhaps, would clearly not avoid exclusion from patentability if the equivalent claim to a computer program did not. It noted that the alleged divergence identified in the referral did not simply relate to the form of words chosen. The discussion also referred to "the function of the computer program (does the claimed program have technical character) rather than the manner in which it is claimed (e.g. as a computer program, a computer program product or a computer-implemented method)".

The Enlarged Board of Appeal noted that in T 424/03 the board had arrived at a conclusion which clearly contradicted the position (or rather one of the positions) taken in T 1173/97 (OJ 1999, 609). T 1173/97 stated, "Furthermore, the board is of the opinion that with regard to the exclusions under Art. 52(2) and (3) EPC, it does not make any difference whether a computer program is claimed by itself or as a record on a carrier ...," (point 13 of the Reasons), whereas T 424/03 stated, "The subject-matter of claim 5 has technical character since it relates to a computer-readable medium, i.e. a technical product involving a carrier" (see decision T 258/03, point 5.3 of the Reasons). Thus there was a difference between the positions taken in T 1173/97 and T 424/03 on this point. It was still however necessary to decide whether this difference constitutes a divergence allowing a question to be referred by the President on the point.

The Enlarged Board of Appeal pointed out that the referral did not identify, and they were not aware of, any decision whatsoever of one of the boards of appeal of the EPO which followed T 1173/97 on this point (although T 1173/97 was evidently considered decisive in its definition of "further technical effect" and abandonment of the contribution approach to exclusion). Secondly, the conclusion arrived at in T 424/03 had not been challenged in any later decisions; nor was it isolated but rather came as the last of a series of decisions, the logic of which was consistent and, at least to the knowledge of the members of the Enlarged Board, had also not been challenged in any later decision of a board of appeal of the EPO (national court decisions were another matter).

The Enlarged Board of Appeal came to the conclusion that the position taken in T 424/03 that a claim to a program on a computer-readable storage medium was necessarily not excluded from patentability by the provisions of Art. 52(2) and (3) EPC 1973 was in fact a consequence of the principles laid out in T 1173/97. It stated that T 424/03 did deviate from a view expressed in T 1173/97, concerning whether a claim to a program on a computer-readable medium necessarily avoided exclusion from patentability under Art. 52(2) EPC 1973. However, this was a legitimate development of the case law. Therefore, the Enlarged Board of Appeal decided that there was no divergence which would make the referral of this point to the Enlarged Board of Appeal by the President admissible (for further legal considerations which were taken into account in that regard, see points 5 to 7 of Opinion G 3/08).